Spatial Data Quality Assessment Report

This report presents the evaluation of positional accuracy for both horizontal and vertical datasets in the study area of Balzan, Malta. The analysis focuses on identifying data inconsistencies, evaluating accuracy levels, and assessing the reliability of the datasets.

Methodology

The positional accuracy assessment was performed by comparing coordinates from the test dataset with an independent reference dataset. For the horizontal analysis, differences in X and Y coordinates were computed for each point, followed by the calculation of squared errors and combined positional error values. For the vertical analysis, elevation differences (ΔZ) were calculated and squared.

The Root Mean Square Error (RMSE) was used to quantify the overall positional accuracy. Accuracy values were then derived at the 95% confidence level using standard statistical conversion factors.

Outlier detection was carried out using a rejection threshold based on error magnitude. Points exceeding this threshold were classified as outliers and excluded from the final accuracy computation to ensure a representative evaluation of the dataset.

Horizontal Accuracy

The horizontal dataset exhibits a high level of consistency, with most positional differences remaining within a very small range. The calculated RMSE is **0.2038 m**, resulting in a horizontal accuracy of **0.3528 m at the 95% confidence level**.

A rejection level of **0.611** was applied to identify and remove extreme outliers. During the analysis, one point was identified as an outlier due to an exceptionally large positional error compared to the rest of the dataset. While the majority of points showed very small squared errors (generally below 0.1), this particular point presented an extremely large deviation (error value ≈ **64**), clearly inconsistent with the overall distribution.

After removing this outlier, the dataset shows a stable and reliable pattern, confirming that the horizontal positioning is highly accurate and suitable for precise spatial applications.

Vertical Accuracy

The vertical dataset shows significantly higher variability compared to the horizontal case. The computed RMSE is **5.706 m**, corresponding to a vertical accuracy of **11.184 m at the 95% confidence level**.

Most points exhibit moderate elevation differences; however, several observations show substantially larger errors. In particular, three points were identified as outliers due to their large squared residuals:

- One point with $\Delta Z^2 \approx$ **151** ($\Delta Z \approx 12.3$ m)
- One point with $\Delta Z^2 \approx$ **334.9** ($\Delta Z \approx 18.3$ m)
- One point with $\Delta Z^2 \approx$ **625.8** ($\Delta Z \approx 25.0$ m)

These values are significantly higher than the rest of the dataset and clearly deviate from the general trend. After excluding these points, the remaining data shows a more consistent distribution, although still less precise than the horizontal dataset.

The higher variability in vertical accuracy reflects limitations in elevation data, likely due to interpolation processes, terrain complexity, and resolution constraints associated with Digital Terrain Models.

Outlier Analysis

The identified outliers represent clear anomalies rather than natural variability. Their inclusion would have significantly inflated the RMSE and led to an overestimation of the dataset's error.

In the horizontal dataset, only **1 out of 40 points (2.5%)** was removed, indicating very high data consistency. In contrast, **3 out of 31 points (9.68%)** were removed from the vertical dataset, reflecting greater variability in elevation data.

Final RMSE and accuracy values were computed after excluding the identified outliers to ensure a reliable representation of the dataset quality.

Discussion

A clear distinction exists between horizontal and vertical accuracy. Horizontal positioning is highly precise, with errors generally at the centimeter level, while vertical accuracy is significantly lower, with errors reaching several meters.

This difference is expected due to the nature of data acquisition. Horizontal coordinates are typically measured with higher precision, whereas vertical values derived from Digital Terrain Models are affected by interpolation, terrain complexity, and resolution limitations.

The relatively higher vertical errors suggest that the dataset may not be suitable for applications requiring high elevation precision. Despite this, the dataset overall demonstrates good quality, particularly in the horizontal component.

Metadata Assessment

The metadata associated with the datasets was reviewed to assess completeness and usability. Basic information such as dataset structure and coordinate values is available; however, detailed documentation regarding data acquisition methods, processing steps, and update history is limited.

This lack of complete metadata may affect the interpretability and long-term usability of the dataset, highlighting the importance of proper metadata documentation in spatial data management.

Attribute Accuracy

A visual inspection of attribute data was conducted to identify inconsistencies between spatial features and their associated attributes. No major attribute mismatches were observed; however, minor inconsistencies may occur due to classification or labeling differences.

This suggests that the attribute data is generally reliable, although careful validation is recommended for critical applications.

Logical Consistency

Logical consistency refers to the internal consistency of spatial data, including the absence of gaps, overlaps, or invalid geometries. While a full topology validation was not performed, no evident inconsistencies were observed during the analysis.

This indicates that the dataset maintains an acceptable level of structural integrity.

Temporal Accuracy

Temporal accuracy refers to how up-to-date and time-relevant the data is. Inaccuracies in temporal data can significantly impact decision-making. For example, outdated elevation data may misrepresent current terrain conditions, and outdated land-use data may lead to incorrect planning decisions.

These examples highlight the importance of maintaining up-to-date datasets in spatial data applications.

Conclusion

The results indicate that the horizontal dataset provides **high positional accuracy (0.35 m)** and is suitable for high-precision applications. The vertical dataset, with an accuracy of approximately **11.18 m**, is less precise but still acceptable for general use.

Outlier detection proved essential in obtaining reliable results. By removing a small number of anomalous points, the analysis was able to provide a realistic representation of data quality.

The results highlight the importance of quality assessment and outlier handling in ensuring reliable spatial data for decision-making.

Overall, the dataset is of good quality, with the main limitations related to vertical accuracy and a small number of extreme observations.